When Portfolio Rewards Miss the Target: Exact Mismatch and Policy Alignment

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Abstract

A risk-sensitive portfolio reward need not implement the investor's stated objective. We examine this difference in two settings. For independent returns, we derive the exact leading policy discrepancy that remains after optimally calibrating a quadratic wealth-increment reward: a common scalar penalty cannot recover terminal mean–variance equilibrium when interest is nonzero and the horizon exceeds one period. For constrained allocation with turnover costs, the mean–variance auxiliary identity shows why risk and cost coefficients must be calibrated together. A walk-forward study of ten US industry portfolios over 432 months confirms fitted-policy recovery across 17 settings. Scalar calibration and a log-minus-variance reward retain allocation differences, while an adapted mean–variance policy iteration method matches the reference. Better alignment does not yield a statistically clear realised performance gain. The analysis separates the economic meaning of a reward from optimisation accuracy and investment performance, allowing each to be checked directly.

CCS Concepts

• **Computing methodologies** → **Reinforcement learning**; • **Applied computing** → *Economics*.

Keywords

portfolio allocation, reward alignment, mean–variance, reinforcement learning, transaction costs

1 Introduction

Suppose an investor asks for mean return minus a variance penalty. Replacing that preference with return minus a squared-return penalty looks harmless, but it also penalises the square of the mean. A learner can optimise the resulting reward accurately and still choose the wrong policy for the stated preference. Interest, trading constraints and transaction costs make the distinction harder to resolve by tuning one coefficient.

Financial learning uses several legitimate objectives. Jiang et al. [10] combine log return and variance; Simon et al. [12] optimise investor utility directly; Cong et al. [4] allow whole-trajectory objectives. Their coexistence makes the reference objective essential: policy disagreement is meaningful only after the intended preference has been specified.

We study the map from reward to policy. First, an independent-return benchmark gives an exact expression for the discrepancy that remains after the best scalar calibration of a wealth-increment reward. Second, a constrained historical study shows how a risk adjustment also changes the effective turnover penalty. Joint calibration restores the fitted preference through the established mean auxiliary used in MVPI [13]. The historical test then asks a separate question: does this alignment improve realised outcomes?

The answer is instructive. Calibration recovers the fitted policy, yet the main realised MV differences are not statistically clear. A useful explanation of a portfolio learner should therefore identify its induced preference as well as its prediction or backtest performance.

2 Related work and comparison design

Objective-aware financial learning. Simon et al. [12] use deep parametric portfolio policies and US equity characteristics. Their Section 4.1, Eq. (17), specifies centred mean–variance utility; Section 3.6, Eqs. (14)–(16), treats turnover costs. Their Table 4 studies alternative preferences. This is a positive example of objective-aware learning, and our fitted covariance-MV reference shares this preference, not their architecture or empirical universe. Gollart and Okhrin [8], Section 4.2.1, Table 1, use zero intermediate rewards and terminal wealth utility, with modified DDPG tested against an analytical Heston–Nandi benchmark. Known-policy validation therefore predates our work. Chaouki et al. [3], Section 3, likewise compare learned and known controls.

Additive rewards and trajectory objectives. Jiang et al. [10], Eq. (8), combine net log return with a variance penalty. Eq. (19) adds soft constraint penalties, and their DPG architecture uses CNN/WaveNet representations. Their Table 2 compares investment horizons across US and Canadian stock universes. We adapt the log-minus-variance preference to the same monthly estimates and hard simplex used by every comparator. We use a first-order turnover charge instead of their log cost factor. The common environment isolates the effect of the reward; the architecture and data differ from those in their study. Jiang et al. [9], Eq. (4), penalise variance and trading costs around a portfolio-*value* term, rather than an excess-return increment. Their TD3/PPO/DDPG comparisons motivate studying risk and cost coefficients together, but are not reproduced here. Cong et al. [4], Eqs. (19)–(20), optimise differentiable trajectory objectives, including a window Sharpe ratio. Such objectives are not presumed equivalent to terminal MV.

Mean–variance control and transformations. Precommitment [11] and equilibrium [1, 6] solve different time-inconsistent control questions. The distinction cannot be settled by a backtest ranking. Czichowsky [5], Lemma 3.2 of the inspected preprint, already gives a discrete continuation-covariance hedge. Björk et al. [2], Propositions 8.1 and 10.2 of the long 2016 preprint, give equivalent control objectives and terminal-unit variance penalisation. Mean–variance policy iteration (MVPI) [13] uses an auxiliary-variable transformation that contains our zero-interest calibration. Our question is more specific: how much policy discrepancy remains after the best scalar calibration of the raw increment reward? The next section gives its exact leading coefficient.

3 An exact test of terminal-objective alignment

3.1 Market, reference and discrepancy

Consider independent excess returns $P_t \in \mathbb{R}^n$ with constant mean $m \neq 0$ and positive definite covariance Σ. Wealth evolves as

$$X_{t+1} = sX_t + P_t^\top u_t, \qquad X_0 = x_0, \tag{1}$$

with unrestricted risky dollar positions u_t and constant gross risk-free return $s > 0$. The terminal objective is $J = \mathbb{E}X_T - \varphi \operatorname{Var} X_T$, $\varphi > 0$. The time-consistent equilibrium policy is

$$u_t^{\mathrm{eq}} = \frac{\Sigma^{-1}m}{2\varphi H_t}, \qquad H_t = s^{T-t-1}. \tag{2}$$

It is not the initial-date precommitment maximiser. Define $M = \Sigma + mm^\top$, $d = M^{-1}m$, $v = m^\top d < 1$ and $b = [2\varphi(1-v)]^{-1}$. The reward family is $r_t = \Delta X_t - \kappa(\Delta X_t)^2$, with one constant $\kappa > 0$. Let π^κ be its exact additive-reward optimum, and measure

$$D^2 = \sum_{t=0}^{T-1} \mathbb{E}_{\pi^\kappa} \|\pi^\kappa(t, X_t) - u_t^{\mathrm{eq}}\|^2. \tag{3}$$

This is an occupancy-dependent discrepancy, not a symmetric policy metric.

3.2 What scalar calibration cannot repair

Theorem 1. *For fixed $T \geq 2$, as $s \to 1$,*

$$\min_{\kappa>0} D = c_T|s-1| + O((s-1)^2), \tag{4}$$

$$c_T = \|d\| b \sqrt{\frac{T(T^2-1)}{3} + \frac{v(1-v)T(T-1)}{2}}. \tag{5}$$

At $s = 1$ exact matching holds for $\kappa = \varphi(1-v)$. For $s \neq 1$, $T \geq 2$, exact matching along the surrogate occupancy is impossible.

Proof. Write the Bellman value $W_t(x) = p_t x^2 + q_t x + c_t$, with $p_T = q_T = 0$ and $g_t = \kappa(s-1) - sp_{t+1}$. Completing the square gives $u_t = \alpha_t x + \beta_t$ and

$$\alpha_t = -\frac{dg_t}{\kappa - p_{t+1}}, \qquad \beta_t = \frac{d(1+q_{t+1})}{2(\kappa - p_{t+1})}, \tag{6}$$

$$p_t = \frac{\kappa p_{t+1} - (1-v)g_t^2}{\kappa - p_{t+1}}, \quad 1 + q_t = (1+q_{t+1})\left(s - \frac{vg_t}{\kappa - p_{t+1}}\right). \tag{7}$$

Induction gives $p_t \leq 0$, so the action Hessian is negative definite and these are global Markov optima. Set $e = s-1$, $h = (2\kappa)^{-1}$. The ratios p_t/κ and q_t are independent of κ, hence pathwise $X_t = L_t(e)x_0 + hZ_t(e)$. Consequently $D^2 = F(e,h)$ is exactly quadratic in h, with analytic coefficients near zero and $F(0,h) = T\|d\|^2(h-b)^2$. Its global minimiser is analytic and remains positive near zero.

Put $\kappa = \varphi(1-v)(1+we)$. The leading action difference is de times

$$-X_t + b[(T-t-1)(2-v) - w].$$

At $e = 0$, $\mathbb{E}X_t = x_0 + tvb$ and $\operatorname{Var} X_t = tb^2v(1-v)$. Minimisation yields $w^* = (T-1)(1-v) - x_0/b$. The centred means are $b(T-1-2t)$; summing their squares and variances gives the displayed c_T^2. Analyticity then gives the stated remainder after taking square roots.

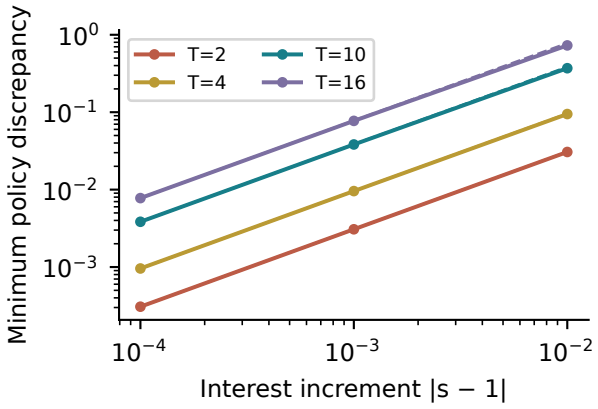


Figure 1: Exact minimum discrepancy (points and solid lines) and local prediction $c_T|s-1|$ (dashed lines). Here $x_0 = \varphi = 1$, $m = (.06, .04, .02)$, volatilities $(.18, .12, .08)$, and correlations $(\rho_{12}, \rho_{13}, \rho_{23}) = (.30, -.10, .05)$. This concerns terminal equilibrium, not the fitted historical reference.

Finally, matching at the first date forces a nonzero equilibrium action and positive subsequent wealth variance. At the final date the surrogate has slope $-d(s-1)$ whereas equilibrium is constant. Almost-sure equality is impossible when $s \neq 1$. □

The coefficient grows as $T^{3/2}$ for large T, but the expansion is for fixed T; no uniform joint interest–horizon limit is claimed. The obstruction concerns this reward family, not all additive objectives.

3.3 A reference-preserving reward

Allow moments to vary by date while returns remain independent. Define $Y_t = H_t P_t^\top u_t$ and use

$$r_t^{\mathrm{design}} = Y_t - \varphi(1-v_t)Y_t^2. \tag{8}$$

The expected reward is strictly concave. Its maximiser is

$$u_t = \frac{\Sigma_t^{-1} m_t}{2\varphi H_t}.$$

Future rewards admit state-independent maximisers, so backward induction recovers the equilibrium. This construction specialises the equivalent-objective approach of Björk et al. [2] and changes the optimum of the original raw reward. Dependent returns require continuation terms, so Eq. (8) is not applied as a historical multiperiod theorem.

4 Constrained alignment with turnover costs

Historical implementation adds constraints and costs. On a fixed past-only estimated state, let C be the simplex, p the pre-trade allocation and $c(u) = \|u - p\|_1$. Define the fitted one-period preference

$$F(u) = m^\top u - \varphi u^\top \Sigma u - \lambda c(u), \quad u \in C. \tag{9}$$

Its unique optimum u^* is a *fitted reference*, not the unknown true market optimum. This section does not extend Theorem 1 to estimated or dependent markets.

Proposition 1. *Let $y^* = m^\top u^*$ and $A = 1 + 2\varphi y^* > 0$. The second-moment reward objective*

$$Q(u) = m^\top u - \frac{\varphi}{A} u^\top M u - \frac{\lambda}{A} c(u) \tag{10}$$

has the same unique maximiser as F.

Proof. For all feasible u,

$$AQ(u) = F(u) - \varphi(m^\top u - y^*)^2 + \varphi(y^*)^2. \tag{11}$$

At u^* the square vanishes and F is maximal. Since $A > 0$, the assertion follows. □

The argument applies to any feasible set with a global maximiser. Convexity and positive definite covariance ensure uniqueness here. The identity follows the mean auxiliary used in MVPI [13].

If only the risk coefficient is divided by A, while the cost coefficient stays λ, the right-hand side of Eq. (11) gains the extra term

$$-2\varphi y^* \lambda c(u). \tag{12}$$

When $y^* > 0$, this increases the effective cost preference. It need not imply monotone realised turnover on the strategy's evolving path. The unconstrained formula $\varphi(1 - \nu)$ likewise need not calibrate a constrained optimum.

Joint calibration uses u^* and therefore serves as a certificate of alignment with a known reference. As a constructive prior-art comparator, we also alternate maximising

$$(1 + 2\varphi y) m^\top u - \varphi u^\top M u - \lambda c(u) \tag{13}$$

and setting $y = m^\top u$. This is an adaptation of the MVPI mean auxiliary to the constrained one-period problem. Including the term $-\varphi y^2$ gives a jointly strictly concave objective, since its quadratic block has Schur complement $\Sigma \succ 0$. Exact block updates converge to the same target. This provides a constructive comparator with the same optimum.

5 Historical study

5.1 Data and protocol

We use Kenneth French's ten value-weighted US industry portfolio monthly total returns and the Fama–French risk-free series [7]. The archived downloads have a recorded SHA-256 manifest. Estimation data start January 1970; evaluation covers January 1990–December 2025 (432 months). This is a retrospective study on the August 2026 revised vintage, not a point-in-time constituent dataset. Industry portfolios are research portfolios, not directly tradeable funds; their internal reconstruction costs are not included. The provider's CRSP processing and risk-free source changes also form part of this data vintage.

For month t, use only the previous 120 months of excess returns to estimate m and the unbiased covariance, shrunk 10% toward its diagonal. All strategies are long-only and fully invested. The primary settings are $\varphi = 5$ and $\lambda = 0.001$ (10 basis points per unit of L1 turnover). Pre-trade weights drift with realised gross asset returns. Net portfolio return uses the explicit first-order convention $u_t^\top R_t - \lambda \|u_t - p_t\|_1$, not a market-impact model or exact intramonth execution equation. The experimental protocol fixed these choices before strategy outcomes were inspected.

5.2 Comparators and measurements

We solve the covariance-MV target, raw second moment ($\kappa = \varphi$), unconstrained scalar calibration ($\kappa = \varphi(1 - \nu)$), reference-based risk-only calibration, joint calibration, and MVPI adaptation. Equal weight and minimum variance provide conventional portfolio controls. Log utility and log-minus-variance adapt the reward preference in Jiang et al. [10]; every method uses the same monthly data and feasible set. The comparison tests induced preferences under common estimates, not which published learning architecture is best.

For alignment, all methods face the target strategy's *same pre-trade state*. We report Euclidean allocation distance and fitted regret $F(u^*) - F(u)$. Each strategy also follows its own pre-trade states for realised performance. This avoids conflating state divergence with the local reward comparison. Quadratic solutions have independent linear-programme first-order certificates; the largest saved common-state certificate is 4.27×10^{-9} in objective units. The observed agreement of calibrated policies is considerably tighter than the acceptance tolerance.

Realised metrics are monthly excess-return MV, annualised excess Sharpe, return, volatility, turnover and drawdown. Uncertainty for paired MV differences uses 1,000 circular moving-block bootstrap resamples, blocks of 12 months, seed 8128. These intervals describe the observed strategy-return sequences, not full estimator-refitting or model-selection uncertainty. Robustness spans 17 prespecified combinations: $\varphi \in \{2, 5, 10\}$ and cost $\{0, 10, 25\}$ bp at the primary estimator, plus windows $\{60, 120, 240\}$ and covariance shrinkage $\{0, 0.1, 0.5\}$ at the primary preferences. The overlapping primary case is counted once.

5.3 Results

Table 1 shows that raw second-moment optimisation has mean allocation distance 0.00843. Unconstrained scalar calibration reduces this to 0.00540; reference-based risk-only calibration retains 0.00510. Joint calibration and MVPI agree with the fitted reference to numerical precision. The adapted log-minus-variance objective has distance 0.01286; log utility alone is much farther away. The differences measure departure from the specified MV preference; log utility represents a different investment objective.

Across all 17 settings, the largest mean joint-calibration distance is 1.54×10^{-7}; raw mean distances range from 0.00157 to 0.03682. Thus constraints and trading costs do not break the joint identity, while unconstrained scalar calibration is not a universal repair. Figure 2 relates alignment to historical outcomes, and Figure 3 gives robustness and uncertainty.

Joint calibration has realised monthly MV lower than raw by 1.315 bp, with 95% interval $[-3.318, 0.343]$ bp. Against scalar calibration the difference is 0.385 bp. Its interval is $[-0.570, 1.351]$ bp. Against risk-only calibration the difference is -0.131 bp, with interval $[-0.889, 0.634]$ bp. All include zero. Minimum variance has both a better realised MV score and higher Sharpe than the fitted target in the primary sample. This illustrates estimation risk: precise implementation of estimated preferences cannot guarantee the best realised allocation. Equal weight has higher Sharpe than the target but lower realised MV, so Sharpe ranking alone does not answer even the ex-post MV comparison.

Table 1: Reward-formulation comparison on historical data, 1990–2025. Published rewards are adapted to a common environment; complete published algorithms are not reproduced. Bold identifies our joint-calibration construction, not the best realised performance. Distance and fitted regret use common states; realised metrics use each strategy's own path. Gap and MV score are monthly basis-point units. Calibration improves model alignment; the table does not establish superior investment performance.

Method	Source / role	Allocation distance	Fitted gap (bp)	Excess Sharpe	Realised MV (bp)
Target	Reference objective	0.00000	0.0000	0.570	-8.497
Raw second moment	Uncalibrated baseline	0.00843	0.0148	0.573	-7.182
Unconstrained scalar	Analytical baseline	0.00540	0.0043	0.568	-8.882
Risk-only calibration	Calibration ablation	0.00510	0.0033	0.571	-8.365
MVPI adaptation	Zhang et al. (2021) [13]	0.00000	0.0000	0.570	-8.497
Log minus variance	Jiang et al. (2025) [10]	0.01286	0.0287	0.572	-7.123
Log utility	Log-utility control	0.54848	39.7855	0.344	-133.295
Equal weight	Standard control	0.43495	46.2626	0.622	-11.893
Minimum variance	Standard control	0.23843	9.0252	0.645	5.274
Joint calibration (ours)	**This study**	**0.00000**	**0.0000**	**0.570**	**-8.497**

5.4 Separating optimisation from reward design

A finite-budget experiment assigns independent softmax policy logits to each of the 432 historical common states. Adam uses learning rate 0.05, initial logits $N(0, 0.1^2)$, five matched seeds and 50, 250, 1,000 and 5,000 gradient steps. Each objective uses only its state's lagged estimates. Independent logits isolate optimisation error on each fitted state. They do not test generalisation across states, as a predictive neural policy would.

Let u_r maximise the specified reward and $\hat{u}_r$ be the finite-budget allocation. We report the nonnegative own-reward optimisation gap separately from

$$F(u^*) - F(\hat{u}_r) = [F(u^*) - F(u_r)] + [F(u_r) - F(\hat{u}_r)]. \tag{14}$$

The second bracket is a *signed* target-value difference; it is not the own-reward optimisation gap. At 5,000 steps, mean own-reward optimisation gaps remain approximately 4×10^{-6}, exceeding the exact raw design gap of 1.48×10^{-6}. Thus finite-budget error can dominate the reward discrepancy. Figure 4 shows why these must be kept separate: improving the raw-reward optimiser does not remove its exact design gap. The target and joint reward have the same limiting optimum but can exhibit different finite-budget trajectories because their objective geometries differ.

6 Reconstructing the historical experiment

6.1 Information available at a decision

Write $R_t \in \mathbb{R}^{10}$ for the vector of monthly total returns and $r_{f,t}$ for the risk-free return. For a window of length L, the information used at month t is $\{R_s, r_{f,s} : t - L \le s < t\}$. In particular, neither R_t nor a full-sample covariance enters the allocation. Let $X_s = R_s - r_{f,s}\mathbf{1}$. The sample mean and covariance are

$$m_t = L^{-1} \sum_{s=t-L}^{t-1} X_s, \tag{15}$$

$$S_t = (L-1)^{-1} \sum_{s=t-L}^{t-1} (X_s - m_t)(X_s - m_t)^\top. \tag{16}$$

The implemented covariance is $\Sigma_t = (1-h)S_t + h\,\mathrm{diag}(S_t)$, with $L = 120$ and $h = 0.1$ in the primary experiment. The mean is not shrunk. Thus every reward sees identical estimates; differences are not attributable to different forecasting models. Positive definiteness is checked before optimisation. Historical dependence can be present in the observed returns without changing the algebra on a fixed estimated state. It does change how much confidence one should place in that estimated state as a forecast.

The raw files contain percent returns, which are divided by 100 before any objective is evaluated. Monthly records are aligned by date; missing-value sentinels, duplicate dates and missing joined observations are rejected. Estimation data begin in 1970 to support the longest 240-month window at the first 1990 decision. Observations after December 2025 are excluded even though the retrieved files extend further. The saved date range therefore fixes the comparison independently of the most recent available month.

6.2 Two uses of the same historical month

The alignment calculation and the investment calculation answer different questions. For alignment, all rewards receive the reference strategy's pre-trade portfolio p_t^*. Define u_t^* as the fitted target allocation and u_t^r as the allocation induced by reward r on that common state. We record

$$d_t^r = \|u_t^r - u_t^*\|_2, \qquad g_t^r = F_t(u_t^*) - F_t(u_t^r). \tag{17}$$

The first measures the change in portfolio weights, while the second measures the economic cost under the stated fitted preference. A small objective gap can accompany a visible weight difference

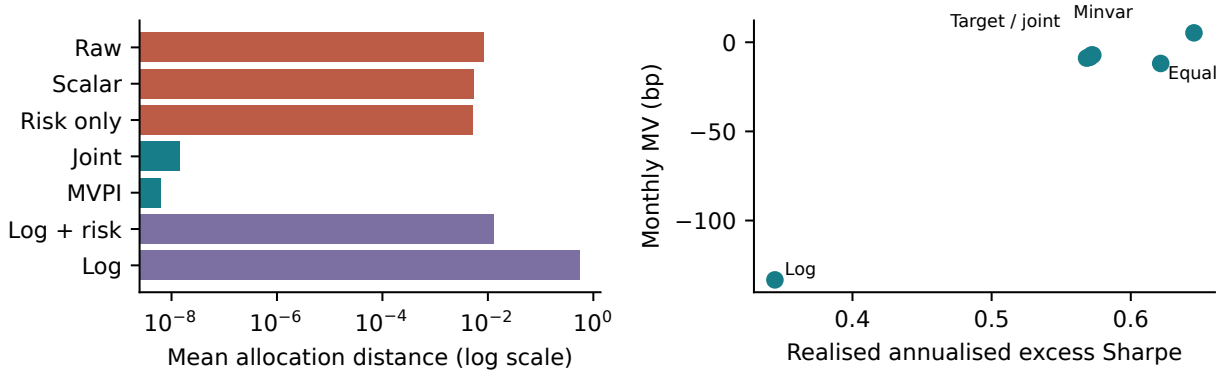


Figure 2: Left: mean common-state allocation distance; the logarithmic axis exposes numerical agreement for joint calibration and MVPI. Right: realised Sharpe and monthly MV need not rank policies alike. Equal weight has higher Sharpe but lower MV than the fitted target.

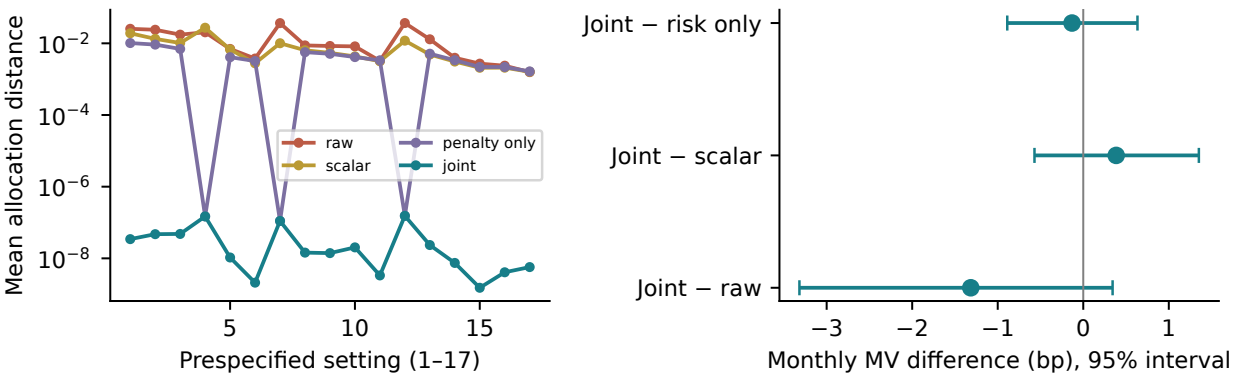


Figure 3: Left: alignment across the 17 prespecified settings, ordered by window, risk penalty, cost and shrinkage. Right: paired 12-month-block bootstrap intervals all include zero. Exact fitted alignment does not imply a statistically clear realised MV advantage.

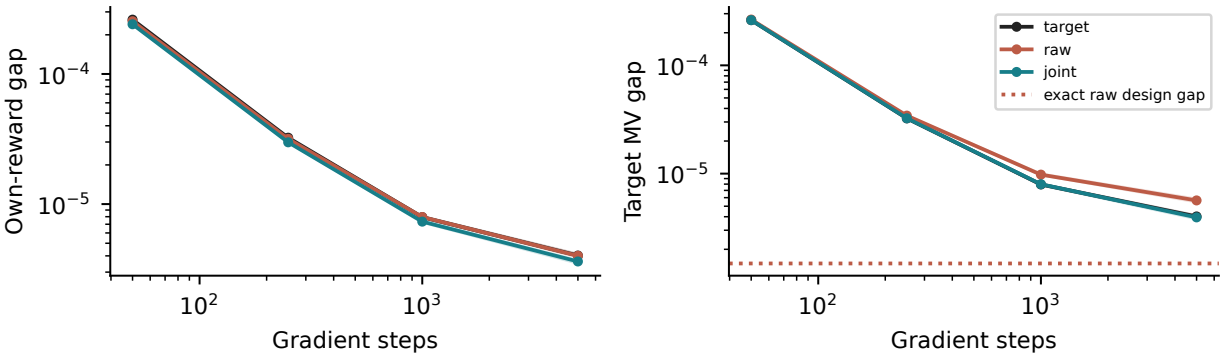


Figure 4: Finite-budget softmax optimisation on historical fitted states. Lines are five-seed means, shading the seed range. Left uses each method's own reward; right uses the common MV target. The dotted line is the exact raw reward-design gap. This diagnostic measures optimisation accuracy, not generalisation.

when the objective is locally flat. Neither quantity uses the month's subsequent realised return.

For investment performance, each method instead carries its own pre-trade portfolio p_t^r. It solves its reward again on that state, giving $\vec{u}_t^r$. The recorded turnover and net return are

$$T_t^r = \|\vec{u}_t^r - p_t^r\|_1, \qquad r_t^r = (\vec{u}_t^r)^\top R_t - \lambda T_t^r. \tag{18}$$

All methods start from equal weights. Before the next decision, holdings drift according to

$$p_{t+1,i}^r = \frac{\vec{u}_{t,i}^r(1 + R_{t,i})}{1 + (\vec{u}_t^r)^\top R_t}. \tag{19}$$

This uses gross asset returns to update relative holdings and subtracts costs separately in the first-order portfolio-return approximation. Charging exact self-financing costs would require a different wealth-accounting model. Using each method's own holdings in the alignment calculation would mix reward disagreement with the consequences of earlier decisions; the common-state record avoids that confounding.

6.3 What the published baselines represent

The MVPI comparator implements the mean auxiliary of Zhang et al. [13] in the constrained one-period setting. Starting from $y = 0$, it alternates a portfolio maximisation with the update $y \leftarrow m^\top u$. The portfolio step retains the original turnover coefficient. The procedure stops when the change in y is below 10^{-10}, with a maximum of 200 iterations. It computes the reference without first inserting the reference allocation into the reward. Joint calibration, by contrast, uses the already computed reference mean to demonstrate exact alignment. Their numerical agreement is a consistency check between two constructions, not an advantage over MVPI.

The Jiang et al. [10] adaptation maximises

$$\frac{1}{L}\sum_{s=t-L}^{t-1} \log(1 + u^\top R_s) - \varphi u^\top \Sigma_t u - \lambda\|u - p_t\|_1. \tag{20}$$

Setting its variance coefficient to zero gives the log-utility comparator. Logarithmic utility uses total returns, while the fitted MV objective uses excess returns. Both formulations receive the same historical window and feasible set. This is a deliberate comparison of preferences: logarithmic utility is not an algebraic approximation to the specified MV target. Hard simplex constraints and the linear turnover charge replace the source paper's soft restrictions and log cost factor. Its published returns are not pooled with ours, and its predictive architecture is not evaluated.

Equal weight supplies a fixed-allocation control. The minimum-variance control removes the expected-return term from the fitted target while retaining the risk scaling and turnover charge. It therefore tests whether using the estimated mean helps realised performance under this protocol. Its strong primary outcome is evidence that estimation quality matters even when reward alignment is exact.

Numerical certificates and the paired bootstrap procedure are detailed in Appendix B.

7 Interpreting calibration across settings

7.1 Which comparison can establish improvement?

A calibration rule is evaluated against a stated financial preference. In this study, improvement in alignment means a smaller d_t^r or g_t^r on the same estimated state. It does not mean that the calibrated strategy earns a higher return in the next month. The realised ranking can reverse because the target's estimated mean and covariance are imperfect forecasts. This distinction is especially relevant when the target itself uses a noisy expected-return estimate and a competing policy happens to reduce exposure to that estimate.

The primary distances put the scale of the comparison in context. Raw second moments give mean Euclidean distance 0.00843; unconstrained scalar calibration reduces it to 0.00540; reference-based risk-only calibration gives 0.00510. Joint calibration and the MVPI adaptation agree with the target to numerical precision. The log-minus-variance adaptation has distance 0.01286. These are distances between ten-dimensional weight vectors, not percentages of portfolio wealth lost. Fitted regret supplies the corresponding objective scale: the raw reward gap is only 0.0148 monthly basis points. A measurable allocation difference can therefore be economically small under the fitted preference.

The historical MV score answers a different question. Joint calibration is 1.315 monthly basis points below raw in the primary sample, while its interval spans both signs. That result does not support a claim that the calibrated policy is a superior investment strategy. Nor does it disprove the calibration identity: subsequent realised returns are absent from that identity. Minimum variance's better realised MV and Sharpe suggest that a strong practical implementation would need to address estimation quality alongside reward design.

7.2 What the robustness grid tests

The first block crosses three risk coefficients, $\varphi \in \{2, 5, 10\}$, with three turnover charges, $\lambda \in \{0, 0.001, 0.0025\}$, at the primary estimation window and shrinkage. The second crosses $L \in \{60, 120, 240\}$ with $h \in \{0, 0.1, 0.5\}$ at the primary risk and cost coefficients. Their shared primary setting is counted once, giving 17 distinct specifications. This is a union of two grids rather than the full cross-product of every choice. It probes economically relevant perturbations without pretending to span all market models.

The maximum setting-mean distance for joint calibration is 1.54×10^{-7}, compared with raw mean distances ranging from approximately 0.00157 to 0.03682. This supports the identity across the evaluated covariance estimates, constraints and cost levels. It does not supply 17 independent statistical replications: all settings reuse the same asset universe and overlapping return observations. The robustness runs compare the target, raw, scalar, risk-only, joint, equal-weight and minimum-variance policies. The MVPI and logarithmic comparators are evaluated in the primary study, rather than silently treated as rerun throughout this grid.

At zero transaction cost the risk-only and joint reward definitions coincide. With positive cost, the difference depends on the target expected return through $A = 1 + 2\varphi y^*$. Constraints can

make distinct objectives select the same boundary allocation, so exact agreement at an individual state need not establish equivalence of the rewards at all states. Conversely, a small discrepancy at the primary setting cannot be assumed to remain small when risk, cost or estimation changes.

7.3 Optimisation effort and reward fidelity

The finite-budget comparison assigns independent logits to each historical state. Softmax enforces strictly positive weights at finite logits, so exact boundary allocations can be approached without being attained. This helps explain why a high-accuracy constrained solver and a finite-budget softmax optimiser need not have equal residual errors even on the same objective. All three trained objectives use identical initialisation distributions, seeds and step budgets; the diagnostic compares numerical effort under that common parameterisation.

The own-reward gap measures whether the optimiser solved the reward it was given. The signed transfer term in the target decomposition instead measures how the approximate policy changes the target value relative to the exact reward optimum. It can have either sign. Adding own-reward suboptimality directly to target-unit design regret would mix different objective scales and would not reproduce the total target gap. At 5,000 steps, mean own-reward gaps remain around 4×10^{-6}, larger than the exact raw design gap of 1.48×10^{-6}. Improving the optimiser can therefore be the larger immediate numerical intervention, while calibration addresses the distinct discrepancy that persists at exact optimisation.

8 Implications and scope

The analytical result is proved for independent returns and unrestricted dollar positions. The historical identity is proved for a fitted one-period problem with a positive calibration factor; its numerical application uses constrained allocations and lagged data. Neither result establishes a true historical terminal-MV equilibrium under dependence. The historical returns themselves need not be iid, but the rolling one-period reference does not identify their complete conditional law or continuation hedge.

The practical procedure is to state preferences before training, identify an appropriate reference, solve or approximate the reward optimum, and audit both on identical states. If a second-moment surrogate is used with turnover costs, Eq. (12) identifies a concrete source of preference drift. A reference-based joint calibration or existing auxiliary-variable method removes that fitted discrepancy. Subsequent learning and historical performance remain separate empirical questions.

Scalar tuning has a precise limit in the independent terminal-wealth problem. With constraints and turnover costs, joint calibration restores the fitted one-period preference, while the realised outcome still depends on estimation quality. These results make reward semantics testable before interpreting a learned policy or comparing backtests. Extending the audit to dependent terminal equilibria and point-in-time investable portfolios would require a richer reference model.

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Appendix

A Complete robustness results

Table 2 reports all 17 saved specifications underlying Figure 3. Each row averages Euclidean allocation distance over the same 432 historical months. These are repeated specifications on one dataset, not independent samples. The primary setting is $L = 120$, $\varphi = 5$, $\lambda = 10$ bp and $h = 0.1$. The three columns isolate an uncalibrated reward, calibration of risk alone, and joint calibration of risk and costs.

Table 2: Complete common-state alignment results. L is the estimation window in months, λ is the cost per unit of L1 turnover in basis points, and h is diagonal covariance shrinkage. Distances are weight-vector norms, not returns. Bold identifies joint calibration (ours).

L	φ	λ (bp)	h	Raw distance	Risk-only distance	**Joint (ours)**
60	5	10	0	0.02541	0.01018	$\mathbf{3.44 \times 10^{-8}}$
60	5	10	0.1	0.02391	0.00921	$\mathbf{4.73 \times 10^{-8}}$
60	5	10	0.5	0.01747	0.00698	$\mathbf{4.80 \times 10^{-8}}$
120	2	0	0.1	0.02032	0.00000	$\mathbf{1.48 \times 10^{-7}}$
120	2	10	0.1	0.00697	0.00411	$\mathbf{1.06 \times 10^{-8}}$
120	2	25	0.1	0.00376	0.00320	$\mathbf{2.10 \times 10^{-9}}$
120	5	0	0.1	0.03664	0.00000	$\mathbf{1.11 \times 10^{-7}}$
120	5	10	0	0.00874	0.00566	$\mathbf{1.45 \times 10^{-8}}$
120	5	10	0.1	0.00843	0.00510	$\mathbf{1.40 \times 10^{-8}}$
120	5	10	0.5	0.00822	0.00413	$\mathbf{2.02 \times 10^{-8}}$
120	5	25	0.1	0.00324	0.00335	$\mathbf{3.34 \times 10^{-9}}$
120	10	0	0.1	0.03682	0.00000	$\mathbf{1.54 \times 10^{-7}}$
120	10	10	0.1	0.01296	0.00513	$\mathbf{2.35 \times 10^{-8}}$
120	10	25	0.1	0.00394	0.00338	$\mathbf{7.45 \times 10^{-9}}$
240	5	10	0	0.00271	0.00219	$\mathbf{1.51 \times 10^{-9}}$
240	5	10	0.1	0.00237	0.00218	$\mathbf{4.09 \times 10^{-9}}$
240	5	10	0.5	0.00157	0.00165	$\mathbf{5.71 \times 10^{-9}}$

B Numerical verification and uncertainty

Quadratic objectives are solved with auxiliary variables $z_i \geq |u_i - p_i|$ and simplex constraints. The implemented minimisation is scaled for numerical conditioning, but certificates are computed in the original objective units. Given a feasible iterate x in the extended polytope $\mathcal{P}$, a separate linear programme gives the certificate

$$C(x) = \max_{v \in \mathcal{P}} \nabla f(x)^\top (x - v), \tag{21}$$

where f is the convex negative reward. Convexity implies $f(x) - \min f \leq C(x)$. This checks accuracy independently of the quadratic optimiser's success flag. Poorly certified iterates are retried at different scales and rejected if the certificate remains above 2×10^{-7}. The log comparators use a separate concave optimisation problem; the quadratic certificate is not claimed for them.

Realised monthly MV is the sample mean of net excess returns minus φ times their sample variance. Excess Sharpe annualises the monthly mean-to-standard-deviation ratio by $\sqrt{12}$. These are evaluated on each strategy's complete own-state return path. For paired MV uncertainty, the same circular 12-month blocks are sampled for both strategies, with 1,000 repetitions and seed 8128. Shared blocks retain contemporaneous dependence between strategies and some serial dependence within each path. The intervals condition on the fitted strategies' realised paths: they do not repeat estimation, select hyperparameters or quantify every source of model uncertainty.